

Generative Modeling with Normalizing Flows

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August 10, 2025

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Monge's Optimal Transport Problem

- Goal: Find a transport map $y = y(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that pushes a source density $\rho(x)$ to a target density $\mu(y)$, minimizing the total transportation cost.

Cost function:

$$M(y) = \int_{\mathbb{R}^n} c(x, y(x)) \rho(x) dx$$

Change of variables:

$$\rho(x) = J^y(x) \mu(y(x)), \quad \text{where } J^y(x) = \det(\nabla y(x))$$

Definition: A map $y(x)$ that satisfies the above condition and minimizes $M(y)$ is called an *optimal map*.

Kantorovich Relaxation and Duality

- Monge's formulation requires a one-to-one transport map from x to y .
- Kantorovich relaxed this by allowing mass at y to come from multiple x , and mass at x to split across destinations y .
- This leads to a new optimization over **joint distributions** $\pi(x, y)$, subject to marginal constraints.

Kantorovich primal problem:

$$\min_{\pi} \int c(x, y) \pi(x, y) dx dy$$

subject to:

$$\int \pi(x, y) dy = \rho(x), \quad \int \pi(x, y) dx = \mu(y)$$

Dual problem:

$$\max_{u, v} \left[\int u(x) \rho(x) dx + \int v(y) \mu(y) dy \right]$$

subject to:

$$u(x) + v(y) \leq c(x, y) \quad \text{for all } x, y$$

Brenier's Theorem

- If ρ and μ are two absolutely continuous probability densities on $\mathbb{R}^d$ with finite second moments, there exists a **unique** optimal transport map T^* pushing ρ forward to μ .
- This map is the gradient of a convex potential function ϕ :

$$T^*(\mathbf{z}) = \nabla\phi(\mathbf{z})$$

- $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, and $\nabla\phi(\mathbf{z})$ tells us where to send the mass originally at $\mathbf{z}$.
- Under **quadratic cost**, the optimal way to transport mass is to follow the gradient of this convex potential.

Mesh-Free Implementation I

- In the case of quadratic cost $c(x, y) = \frac{1}{2}\|x - y\|^2$, Monge and Kantorovich are connected via:

$$y(x) = \nabla u(x)$$

where u is the Kantorovich potential.

- Let $z^k(x)$ denote the transport map at iteration step k , with:

$$z^0(x) = x, \quad z_i^{k+1} = z^k(x_i)$$

- As the flow evolves, the map converges:

$$\lim_{k \rightarrow \infty} z^k(x) = y(x)$$

Mesh-Free Implementation II

- We consider the case where the densities $\rho(x)$ and $\mu(y)$ are only known through samples:

$$x_i \sim \rho(x), \quad y_j \sim \mu(y), \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- Define $\rho^k(z)$ as the pushforward of $\rho(x)$ under $z^k(x)$
- Estimate the dual functional using Monte Carlo:

$$\hat{D}^k(\beta) = \frac{1}{m} \sum_{i=1}^m u_{\beta}^k(z_i^k) + \frac{1}{n} \sum_{j=1}^n (u_{\beta}^k)^*(y_j)$$

Mesh-Free Implementation III

- This estimate approximates the true functional:

$$\tilde{D}^k(\beta) = \int u_{\beta}^k(x) \rho^k(x) dx + \int (u_{\beta}^k)^*(y) \mu(y) dy$$

- Instead of following the full gradient flow, we restrict the maps at each step to a parametrized form:

$$z_i^{k+1} = \nabla u_{\beta}^k(z_i^k)$$

- The potential is:

$$u_{\beta}^k(z) = \frac{1}{2} \|z\|^2 + \sum_l \beta_l F_l^k(z)$$

- Therefore, the gradient of the potential is:

$$\nabla u_{\beta}^k(z) = z + \sum_l \beta_l \nabla F_l^k(z)$$

Computing β via Gradient Descent

- To minimize the dual objective $\hat{D}^k(\beta)$, we compute the gradient with respect to each β_l :

$$\frac{\partial \hat{D}^k(\beta)}{\partial \beta_l} = \frac{1}{m} \sum_{i=1}^m F_l^k(z_i^k) - \frac{1}{n} \sum_{j=1}^n F_l^k(y_j)$$

- Apply a single step of gradient descent:

$$\bar{\beta}_l \leftarrow \bar{\beta}_l - \eta \cdot \frac{\partial \hat{D}^k(\beta)}{\partial \beta_l}$$

- Substituting the expression:

$$\bar{\beta}_l \leftarrow \bar{\beta}_l - \eta \left(\frac{1}{m} \sum_{i=1}^m F_l^k(z_i^k) - \frac{1}{n} \sum_{j=1}^n F_l^k(y_j) \right)$$

- At convergence (as $k \rightarrow \infty$), the residual vanishes:

$$\frac{1}{m} \sum_{i=1}^m F_l^\infty(z_i^\infty) = \frac{1}{n} \sum_{j=1}^n F_l^\infty(y_j)$$

Choosing Basis Functions $F_l(z)$

- To complete the procedure, we must specify the form and number of functions $F_l(z)$ used at each time step.
- Key considerations:
 - Avoid over/under-resolving the densities $\rho^k(z)$ and $\mu(y)$.
 - Functions F_l should be designed to perform well in high-dimensional settings.
- **Example:** A radially symmetric potential:

$$F_l(z) = r \operatorname{erf}\left(\frac{r}{\alpha}\right) + \frac{\alpha}{\sqrt{\pi}} e^{-(\frac{r}{\alpha})^2}, \quad r = \|z - \tilde{z}_l\|$$

- Bandwidth α depends on local densities:

$$\alpha \propto \left(\frac{n_p}{n+m} \left(\frac{1}{\bar{\rho}(\tilde{z}_l)} + \frac{1}{\bar{\mu}(\tilde{z}_l)} \right) \right)^{1/d}$$

Constructing the Transport Map with the RBF

- We restrict the transport map to a parameterized potential:

$$u_{\beta}^k(z) = \frac{1}{2}\|z\|^2 + \sum_I \beta_I F_I^k(z)$$

- Its gradient defines the update step:

$$\nabla u_{\beta}^k(z) = z + \sum_I \beta_I \nabla F_I^k(z)$$

$$z_i^{k+1} = \nabla u_{\beta}^k(z_i^k)$$

- Using a radially symmetric basis $F_I(z) = r \operatorname{erf}(r/\alpha) + \frac{\alpha}{\sqrt{\pi}} e^{-(r/\alpha)^2}$, we obtain the explicit update:

$$z^{k+1} = z^k + \sum_I \beta_I f_I(r)(z^k - \tilde{z}_I)$$

where:

$$f_I(r) = \frac{1}{r} \frac{dF}{dr} = \frac{\operatorname{erf}(r/\alpha)}{r}, \quad r = \|z^k - \tilde{z}_I\|$$

Iterative Viewpoint of the Transport Map

- Start with:

$$z^0(x) = x$$

- Apply successive maps:

$$z^1(x) = \nabla u_\beta^0(z^0(x)) = x + \text{pert}_0$$

$$z^2(x) = \nabla u_\beta^1(z^1(x)) = z^1(x) + \text{pert}_1$$

$$z^3(x) = \nabla u_\beta^2(z^2(x)) = z^2(x) + \text{pert}_2$$

- After k steps:

$$z^k(x) = x + \text{pert}_0 + \text{pert}_1 + \cdots + \text{pert}_{k-1}$$

- Note: In each step, the center $\tilde{z}_l$ of F_l^k is chosen randomly (either from the current particles or from the target distribution) to localize the map.

Gradient as Sum of Convex Functions

- The transport map is defined as:

$$\nabla u_{\beta}^k(z) = z + \sum_I \beta_I \nabla F_I^k(z)$$

which is the gradient of the potential:

$$u_{\beta}^k(z) = \frac{1}{2} \|z\|^2 + \sum_I \beta_I F_I^k(z)$$

- This can be reformulated as:

$$\nabla u_{\beta}^k(z) = \nabla \left(\frac{1}{2} \|z\|^2 + \sum_I \beta_I F_I^k(z) \right)$$

Normalizing Flow Results I

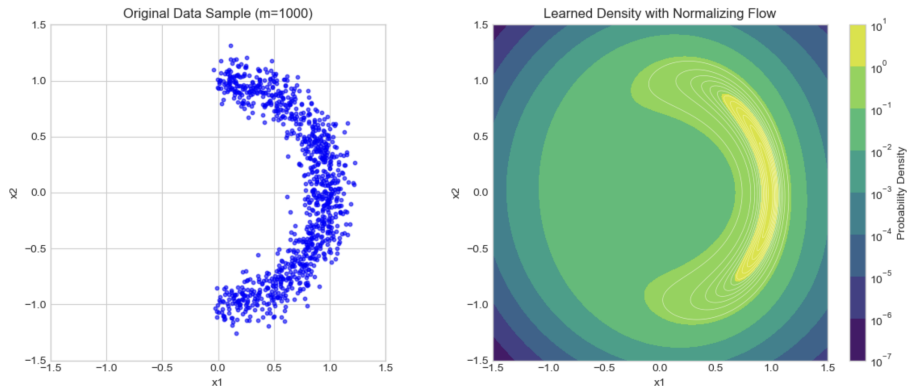


Figure: Results of the Normalizing flow after 1000 steps on a crescent distribution

$$\alpha = \sqrt{2\pi} \left(\Omega_n^{-1} \frac{n_p}{m} \right)^{\frac{1}{n}} e^{\frac{\|x_0\|^2}{2n}}, \quad f(r) = \frac{1}{\alpha} \frac{\text{erf}\left(\frac{r}{\alpha}\right)}{\frac{r}{\alpha}}.$$

Normalizing Flow Results II

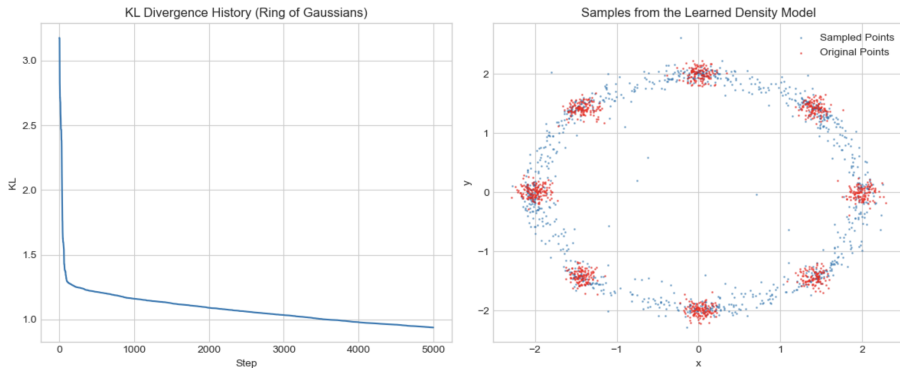


Figure: Results of the Normalizing flow after 5000 steps on a ring distribution

Adaptations to the model

- In order to incorporate the theory of optimal transport, we look back at the equation,

$$\nabla u_{\beta}^k(z) = z + \sum_l \beta_l \nabla F_l^k(z).$$

- We need to maintain convexity!
- To do this, instead of composing over the previous map like in Normalizing Flows, we add maps instead, taking the gradient based on the initial point:

$$z_i^{k+1} = z_i^k + \beta_k \nabla_{z^0} F_l^k(\|z^0 - \tilde{z}^{k+1}\|).$$

We investigated two approaches:

- Convex radial basis functions
- Input Convex Neural Networks (ICNNs)

Convexity Conditions and Brenier's Theorem

- **Positive weights (β):** Convex functions are closed under non-negative linear combinations. The sum remains convex, and its gradient remains a valid transport map.
- **Negative weights:** If any $\beta_l < 0$, convexity may be lost. The resulting sum may no longer be the gradient of a convex function. Careful control of step sizes is needed to maintain structure.
- **Connection to Brenier's Theorem:** For quadratic cost, the optimal transport map is guaranteed to be the gradient of a convex function. Ensuring u_β^k is convex preserves compatibility with Brenier's structure.

Checking Convexity of Updated Potential for Adaptable β_l

- After each update, potential function is

$$u_{\beta}^{k+1}(z) = u_{\beta}^k(z) + \beta_{k+1} F_{k+1}(z)$$

where u_{β}^k and $F_{k+1}(z)$ are convex.

- If $\beta_{k+1} < 0$, $u_{\beta}^{k+1}(z)$ is not guaranteed to be convex.
- We check convexity **empirically** by testing whether the fixed initial sample points

$$(z_0^{(i)}, u_{\beta}^{k+1}(z_0^{(i)}))_{i=1}^m$$

can be interpolated by a convex function.

- This is equivalent to verifying that for each i there exists a subgradient g_i such that:

$$u_{\beta}^{k+1}(z_0^{(j)}) \geq u_{\beta}^{k+1}(z_0^{(i)}) + g_i^{\top} (z_0^{(j)} - z_0^{(i)}) \quad \forall j.$$

- If these inequalities are not feasible, **increase** β_{k+1} (e.g., divide by a small factor > 1) and repeat the check until it passes.

Interpolation Check and Dual LP

- From the interpolation condition:

$$u_{\beta}^{k+1}(z_0^{(j)}) \geq u_{\beta}^{k+1}(z_0^{(i)}) + g_i^{\top} (z_0^{(j)} - z_0^{(i)}) \quad \forall j,$$

we only need to check if these inequalities are **feasible** for g_i .

- By **Farkas' Lemma**, the inequalities are infeasible for some $i \iff$ there exists $\alpha \geq 0$ such that

$$\sum_{j=1}^m \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0, \quad \sum_{j=1}^m \alpha_j (u_{\beta}^{k+1}(z_0^{(j)}) - u_{\beta}^{k+1}(z_0^{(i)})) < 0.$$

- Since LP solvers do not allow strict inequalities, we solve:

$$\min_{\alpha \geq 0} \sum_{j=1}^m \alpha_j (u_{\beta}^{k+1}(z_0^{(j)}) - u_{\beta}^{k+1}(z_0^{(i)})) \quad \text{s.t.} \quad \sum_{j=1}^m \alpha_j (z_0^{(j)} - z_0^{(i)}) = 0.$$

- If the LP is **unbounded** for any $i \Rightarrow$ inequalities infeasible $\Rightarrow$ not convex.
- Using the dual LP greatly reduces runtime when m (number of points) is large.

Input Convex Neural Networks (ICNNs)

- **Goal:** Parameterize a convex function $F(x)$ that is differentiable.
- **Convexity Guarantee:**
 - Constrain certain weights to be non-negative.
 - Use convex, non-decreasing activation functions (e.g., ReLU, softplus).
- **Architecture:**
 - Each hidden layer: $z_{l+1} = \sigma(W_{l+1}^{(z)}z_l + W_{l+1}^{(x)}x + b_{l+1})$
 - $W_{l+1}^{(z)} \geq 0$ ensures convexity in x .
 - Skip connections from input x preserve convexity.
- **Properties:**
 - Universal approximators of convex functions on compact sets.
 - Gradient $\nabla F(x)$ defines a *monotone map*.
 - Suitable for Brenier maps in optimal transport.

Input Convex Neural Network (ICNN)

- **Brenier's Theorem:** For quadratic cost, the optimal transport map T is the gradient of a convex potential ϕ .
- **Parameterization:**

$$\phi_\alpha(x) = \frac{\alpha}{2} \|x\|^2 + \sum_{k=1}^K \beta_k F_k(x), \quad x \in \rho(x)$$

- **Resulting Map:**

$$T(x) = \nabla \phi_\alpha(x) = \alpha x + \sum_{k=1}^K \beta_k \nabla F_k(x)$$

- **Design Choice:** Choose each F_k to be an *Input Convex Neural Network (ICNN)* to ensure convexity.
- **Strong Convexity:** The quadratic term $\frac{\alpha}{2} \|x\|^2$ guarantees strong convexity when $\alpha > 0$, ensuring T is invertible and monotone.

Results using ICNNs

Special Case: $\alpha = 1$: $T(x) = x + \sum_{k=1}^K \beta_k \nabla F_k(x)$, Identity map plus a convex perturbation from the ICNN terms.

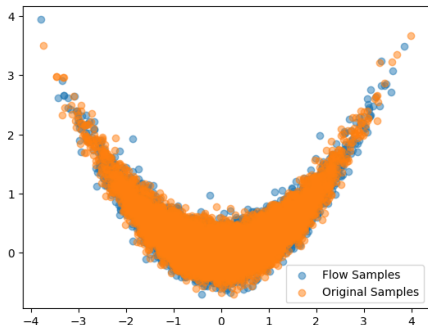


Figure: Banana distribution
 $p(x, y) \propto \mathcal{N}(x; 0, 1) \mathcal{N}\left(y; \frac{x^2}{4}, 0.2\right)$

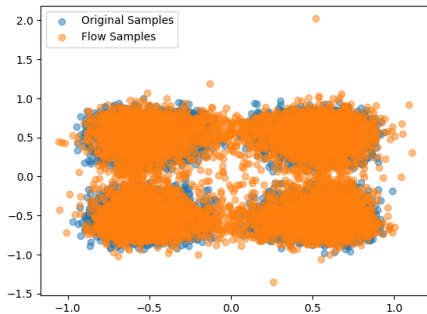


Figure: Convex map $F(x, y) = x^4 + y^4$, visualized by applying ∇F^{-1} to a bivariate normal distribution

Conclusion & Future Work

Conclusion:

- Successfully implemented and debugged a classical non-parametric normalizing flow.
- Developed and tested a modern ICNN-based flow, demonstrating its power and theoretical guarantees.
- Performed ablation studies to validate our model design and understand key hyperparameters.
- The ICNN approach appears more robust and powerful for complex distributions.

Future Work:

- Apply these models to higher-dimensional data (e.g., images).
- Explore the theoretical connections between the two methods.
- Conduct a more extensive hyperparameter search.
- Incorporate AdaBoost and Flow matching techniques

Thank You for Listening!

Special Thanks to Giulio, Nisha, Ricardo, and Tanya!

Questions?

References I

-  E. G. Tabak and C. V. Turner, *A family of non-parametric density estimation algorithms*, *Communications on Pure and Applied Mathematics*, 66(2):145–164, 2013. DOI: 10.1002/cpa.21423.
-  C.-W. Huang, R. T. Q. Chen, C. Tsirigotis, and A. Courville, *Convex Potential Flows: Universal Probability Distributions with Optimal Transport and Convex Optimization*, in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021. Available at arXiv:2012.05942.
-  E. G. Tabak and G. Trigila, *Data Driven Optimal Transport*, *Communications on Pure and Applied Mathematics*, 69(4):613–648, 2016. DOI: 10.1002/cpa.21591. Available at math.nyu.edu.
-  B. Amos, L. Xu, and J. Z. Kolter *Input Convex Neural Networks*, *34th International Conference on Machine Learning, Sydney, Australia, PMLR 70*, 2017. Available at arXiv:1609.07152.